

Consider the optimization problem

$$\min_{x \in \mathbb{R}^2} x_1 + 4x_2$$

subject to

$$\begin{aligned}x_1 + x_2 &= 2, \\x_1 &\geq 0, \\x_2 &\geq 0.\end{aligned}$$

for which the solution is $x^* = (2, 0)^T$ with an optimal value 2.

1. Write a new optimization problem by relaxing the constraint $2 - x_1 - x_2 = 0$, and penalizing its violation with a parameter λ .
2. Solve the new problem for $\lambda = 0, 1$ and 2 , and, for each case, discuss about the value of the objective function at the optimal solution.